BE (E&TC)	PLC SCADA and Automation (Elective-III)	
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AIM: Development of a ladder program for DOL Starter

To develop a ladder logic program Direct-On-Line (DOL) starter in PLC, and verify its operation in both software simulation and hardware.

Apparatus / Software Required: TIA 13(Totally Integrated Automation)

Sr. No.	Equipment / Software	Specification / Example
1	PLC Trainer Kit	Siemens S7-1200
2	PLC Programming Software	TIA Portal (Totally Integrated Automation)
3	Push Button Switches	NO (Start), NC (Stop)
4	Contactor	With 3-phase motor connection
5	Overload Relay	Thermal overload protection
6	Indicator Lamps / LEDs	Motor running indication
7	Power Supply	24V DC for PLC, 3-phase AC for motor

OBJECTIVES

- Understand the working of a DOL starter in motor control.
- Implement start/stop control using PLC ladder logic.
- Ensure motor protection using overload relay logic.

THEORY

A **Direct-On-Line (DOL) starter** is the simplest method of starting a 3-phase induction motor. In PLC-based control:

- **Start button (NO)** energizes the contactor coil.
- **Stop button (NC)** de-energizes the contactor coil.
- **Overload relay (NC)** provides motor protection.

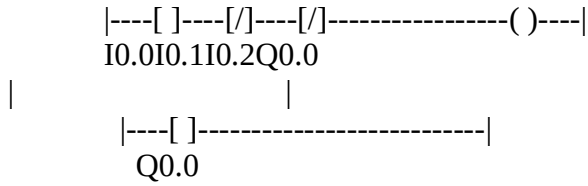
- **Seal-in contact** keeps the motor running after the start button is released.

LADDER LOGIC DIAGRAM

PLC Address Assignment Example

Device	PLC Address
Start Push Button	I0.0
Stop Push Button	I0.1
Overload Relay	I0.2
Contactor Coil / Motor Output	Q0.0

Ladder Diagram Logic



Explanation:

- `I0.0` → Start PB (NO)
- `I0.1` → Stop PB (NC)
- `I0.2` → Overload Relay (NC)
- `Q0.0` → Contactor coil (motor ON)
- The parallel branch with `Q0.0` is the **seal-in contact** for latching.

PROCEDURE

Software Simulation:

1. Create a new PLC project in programming software.
2. Assign input/output addresses.
3. Draw ladder diagram as per above logic.
4. Compile and download the program to PLC/simulator.
5. Test using virtual start/stop buttons.

Hardware Testing:

1. Wire start and stop push buttons to PLC inputs.

2. Connect overload relay NC contact in series with stop PB.
3. Connect PLC output to contactor coil controlling the motor.
4. Run program and test motor start/stop operation.

Observations & Output Table

Start PB	Stop PB	Overload Relay	Motor Output
0	0	0	OFF
1	0	0	ON
0	0	0	ON (Latched)
0	1	0	OFF
0	0	1	OFF

Result

The ladder program for DOL starter was successfully developed and tested. Motor started and stopped as per the control logic, with overload protection.

Applications

- Industrial motor starting systems.
- Conveyor belt systems.
- Pump control systems.

Conclusion

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Questions:

1. What is a Direct-On-Line (DOL) starter and why is it used?
2. List the main components of a DOL starter circuit.
3. Explain the working principle of a DOL starter.

REFERENCES:

[1]Petruzella, F. D. (2017). *Programmable Logic Controllers* (5th ed.). McGraw-Hill Education.
[2] Bolton, W. (2021). *Programmable Logic Controllers* (7th ed.). Elsevier.
[3] Siemens AG. (2023). *TIA Portal Programming Manual*.Siemens Industry Online Support